

Logarithmic cyclicity in Dirichlet-type spaces: the complete phase diagram

Solution packet for arXiv:1301.4375, Problem 3

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Abstract

For

$$D_\alpha = \left\{ f(z) = \sum_{n \geq 0} a_n z^n : \sum_{n \geq 0} (n+1)^\alpha |a_n|^2 < \infty \right\},$$

Problem 3 of Bénéteau–Condori–Liaw–Seco–Sola asks whether $f, \log f \in D_\alpha$ force f to be cyclic. We give the complete answer: the implication holds precisely for $\alpha \geq 0$. The missing positive range follows from the later complete-Pick theorem of Aleman–Richter. For every $\alpha < 0$, a theorem of Dayan–Seco supplies a singular measure on a Beurling–Carleson set whose Fourier coefficients, after a weighted Hölder estimate, give a noncyclic singular inner function S_μ with both S_μ and $\log S_\mu$ in D_α .

1 The question and the answer

A function $f \in D_\alpha$ is *cyclic* when the polynomial multiples of f are dense in D_α , equivalently when there are polynomials p_j such that $p_j f \rightarrow 1$ in D_α . The phrase “ $\log f \in D_\alpha$ ” means that f has an analytic logarithm q on $\mathbb{D}$ and that $q \in D_\alpha$.

The source paper [1] proves the implication for $\alpha > 1$, $\alpha = 0$, and $\alpha = 1$. It obtains a conditional result for the remaining parameters and asks whether its extra approximation hypothesis is necessary. The following settles the whole question.

Theorem 1 (Complete phase diagram). *For a real number α , the following are equivalent:*

1. $\alpha \geq 0$;
2. every f satisfying $f, \log f \in D_\alpha$ is cyclic in D_α .

More precisely, if $\alpha < 0$, there is a singular inner function S_μ such that $S_\mu, \log S_\mu \in D_\alpha$ but S_μ is not cyclic.

Proof intuition. There is a sharp change at the Hardy index. On the nonnegative side, $\log f \in D_\alpha \subset H^1$ forces f to be outer; complete-Pick Smirnov theory then turns the logarithm into a cyclicity certificate. On the negative side, D_α is large enough to contain every inner function. We choose a singular inner function whose singular measure lives on a Beurling–Carleson set, which forces noncyclicity, while arranging just enough Fourier summability of that measure to put the analytic logarithm in D_α .

2 The affirmative half

We first recall the later theorem that closes the interval left open by the source.

Theorem 2 (Aleman–Richter [2]). *Let $D(\nu)$ be a superharmonically weighted Dirichlet space. If $g \in D(\nu)$ is outer and $\log g \in N^+(D(\nu))$, the Pick–Smirnov class of $D(\nu)$, then g is cyclic in $D(\nu)$.*

The standard spaces D_α , $0 \leq \alpha \leq 1$, belong to this family [2, Introduction]. Their normalized complete Nevanlinna–Pick property also gives

$$D_\alpha \subset N^+(D_\alpha). \quad (1)$$

Suppose first that $0 \leq \alpha \leq 1$ and put $q = \log f$. Since $D_\alpha \subset H^2$, we have $q \in H^2 \subset H^1$. The standard analytic-log criterion then says that f is outer. Indeed, radial boundary values satisfy $\log |f^*| = \operatorname{Re} q^*$ almost everywhere, and the H^1 mean-value formula gives

$$\log |f(0)| = \operatorname{Re} q(0) = \int_{\mathbb{T}} \operatorname{Re} q^* dm = \int_{\mathbb{T}} \log |f^*| dm,$$

which is the outer equality (and $f = e^q$ has no zeros). Now $q \in D_\alpha \subset N^+(D_\alpha)$ by (1), so Theorem 2 makes f cyclic.

For completeness, let $\alpha > 1$. The coefficient space D_α is a Banach algebra: it is the one-sided weighted convolution space ℓ_s^2 with $s = \alpha/2 > 1/2$. Hence the exponential power series converges in D_α , and

$$\frac{1}{f} = e^{-q} \in D_\alpha.$$

Choose polynomials $p_j \rightarrow 1/f$ in D_α . Multiplication by $f \in D_\alpha$ is bounded, so $p_j f \rightarrow 1$. This proves cyclicity for all $\alpha \geq 0$.

3 Counterexamples for every negative parameter

We use two facts assembled explicitly in Dayan–Seco [3]. First, the Korenblum–Roberts theorem says that, for $\alpha < 0$, a singular inner function S_μ is cyclic in D_α if and only if

$$\mu(E) = 0 \quad \text{for every Beurling–Carleson set } E \subset \mathbb{T}. \quad (2)$$

Second, their Theorem 6.3 states that for each $p > 2$ there is a nonzero positive singular measure μ , supported on a Beurling–Carleson set, such that

$$(\hat{\mu}(n))_{n \in \mathbb{Z}} \in \ell^p(\mathbb{Z}). \quad (3)$$

Their proof uses a Salem measure and also verifies the Beurling–Carleson property of its support.

Fix $\alpha < 0$. Select $p > 2$ with

$$\frac{\alpha p}{p-2} < -1. \quad (4)$$

If $\alpha \leq -1$, every $p > 2$ works. If $-1 < \alpha < 0$, it is enough to take

$$2 < p < \frac{2}{1+\alpha},$$

an interval which is nonempty precisely because $\alpha < 0$. Take the measure supplied by (3) for this p . Hölder's inequality with conjugate exponents $p/2$ and $p/(p-2)$ yields

$$\sum_{n \geq 1} (n+1)^\alpha |\widehat{\mu}(n)|^2 \leq \left(\sum_{n \geq 1} |\widehat{\mu}(n)|^p \right)^{2/p} \left(\sum_{n \geq 1} (n+1)^{\alpha p/(p-2)} \right)^{(p-2)/p} < \infty, \quad (5)$$

where the final series converges by (4).

Define the associated singular inner function

$$S_\mu(z) = \exp \left\{ - \int_{\mathbb{T}} \frac{\zeta + z}{\zeta - z} d\mu(\zeta) \right\}.$$

Its defining exponent is an analytic logarithm. Expanding the Herglotz kernel gives, up to the choice of Fourier-sign convention,

$$\log S_\mu(z) = -\mu(\mathbb{T}) - 2 \sum_{n \geq 1} \widehat{\mu}(n) z^n. \quad (6)$$

Thus (5) says that $\log S_\mu \in D_\alpha$. Moreover, $S_\mu \in H^\infty \subset H^2$, and $(n+1)^\alpha \leq 1$ shows that $H^2 \subset D_\alpha$. Consequently $S_\mu \in D_\alpha$ as well.

Finally, μ is supported on a Beurling–Carleson set E and is nonzero, so $\mu(E) = \mu(\mathbb{T}) > 0$. Condition (2) fails, and S_μ is not cyclic in D_α . This proves the negative half and completes the proof of Theorem 1. $\square$

4 Audit of the upgrade and its scope

The atomic singular inner function $\exp(-(1+z)/(1-z))$ already gives counterexamples for $\alpha < -1$, because the coefficients of its logarithm have constant magnitude. It does not reach $-1 \leq \alpha < 0$. The substantive upgrade is the choice of p in (4), which converts Dayan–Seco's ℓ^p Fourier measure into the exact weighted ℓ^2 regularity required for every negative α .

As a cross-check, the Salem estimate quoted in their proof is $\widehat{\mu}(n) = O(n^{-s/2+\varepsilon})$ for arbitrary $s \in (0, 1)$, on a support they show to be Beurling–Carleson. Choosing $s - 2\varepsilon > 1 + \alpha$ makes the series in (5) converge directly. This gives the same endpoint coverage without the abstract embedding calculation.

This packet is a literature-assisted synthesis. Aleman–Richter state the theorem used for the positive range. Dayan–Seco state the measure theorem and the Korenblum–Roberts criterion, but do not state the complete D_α phase diagram verbatim. Their 2026 publisher correction only repairs a typeset integral in another part of the proof and does not affect Theorem 6.3 or the argument above [4].

References

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